

General revision I — number and algebra

Directed numbers, fractions, expressions and equations, place value, roots and sequences.

LESSON 191–194

4 × 40 MIN

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S7 WHOLE-COURSE REVIEW

LESSON CLOCK

30'

35'

35'

35'

20'

5

Directed numbers	30 min
Fractions	35 min
Expressions and equations	35 min
Place value, decimals and roots	35 min
Sequences	20 min
Homework	5 min

LEARNING OBJECTIVES

- Calculate with directed numbers in all four operations.
- Divide fractions and add or subtract mixed numbers.
- Simplify an expression and solve a linear equation.
- Work with powers of ten, rounding, roots and the nth term.

TEXTBOOK

National	pp. 545–556
Cambridge	Stage 7, pp. 8–120
Workbook	Checkpoint practice 1

01

Explanation

Directed numbers

CALCULATION	RULE	ANSWER
$-7 + 3$	move right from -7	-4
$-5 - 4$	move left from -5	-9
$3 - (-8)$	subtracting a negative adds	11
$(-6)(-4)$	two negatives multiply to a positive	24
$-20 \div 5$	one negative gives a negative	-4

TWO RULES, NOT MANY

For adding and subtracting, think of movement on the number line. For multiplying and dividing, count the negative signs: an even number of them gives a positive answer, an odd number a negative.

Fractions

CALCULATION	METHOD	ANSWER
$\frac{3}{4} \div 2$	multiply by $\frac{1}{2}$	$\frac{3}{8}$
$5 \div \frac{1}{4}$	how many quarters in 5	20
$\frac{2}{3} \div \frac{4}{9}$	multiply by the reciprocal	$1\frac{1}{2}$
$2\frac{3}{4} + 1\frac{5}{6}$	wholes, then twelfths	$4\frac{7}{12}$
$4\frac{1}{3} - 1\frac{5}{6}$	regroup one whole	$2\frac{1}{2}$

A MIXED NUMBER IS CONVERTED BEFORE DIVIDING, NOT AFTER

$2\frac{1}{4} \div 1\frac{1}{2}$ is $\frac{9}{4} \div \frac{3}{2} = \frac{9}{4} \cdot \frac{2}{3} = \frac{3}{2}$. Turning over a mixed number as it stands gives nonsense.

Expressions and equations

TASK	WORKING	ANSWER
simplify $3x + 5x - 2x$	collect like terms	$6x$
expand $2(3x - 4)$	multiply both terms	$6x - 8$
evaluate $4x - 7$ at $x = 3$	$12 - 7$	5
solve $3x + 7 = 22$	$3x = 15$	$x = 5$
solve $5x - 4 = 3x + 10$	$2x = 14$	$x = 7$
solve $2(x - 3) = 14$	$x - 3 = 7$	$x = 10$

EVERY SOLUTION CAN BE CHECKED IN ONE LINE

Put $x = 7$ back into $5x - 4 = 3x + 10$: $31 = 31 \checkmark$. There is no excuse for a wrong answer to an equation surviving to the end of a paper.

Place value, decimals and roots

TASK	ANSWER	TASK	ANSWER
3.6×10^3	3600	$\sqrt{144}$	12
$4500 \div 10^2$	45	$\sqrt{81}$	9
0.05×10^4	500	$\sqrt[3]{64}$	4
$0.6 + 0.35$	0.95	$\sqrt[3]{125}$	5
2.4×0.5	1.2	$HCF(12, 18)$	6
$6.3 \div 0.7$	9	$LCM(12, 18)$	36
0.8×0.04	0.032	3.847 to 2 d.p.	3.85

COUNT THE DECIMAL PLACES IN A MULTIPLICATION

0.8 has one and 0.04 has two, so the answer has three: 0.032. In a division, make the divisor whole first — $6.3 \div 0.7$ becomes $63 \div 7$.

Sequences

SEQUENCE	DIFFERENCE	NTH TERM	THE 100TH
5, 8, 11, 14	3	$3n + 2$	302
3, 7, 11, 15	4	$4n - 1$	399
2, 9, 16, 23	7	$7n - 5$	695
20, 17, 14, 11	-3	$23 - 3n$	-277

WRITE DN, THEN CORRECT IT

The difference gives the dn ; the first term says what to add or subtract. Test the rule on the second term before using it — five seconds that saves the whole question.

02 Worked examples

EXAMPLE 1 Work out $3 - (-8) + (-6)(-4) \div 8$.

- ① $3 - (-8) = 11$.
Subtracting a negative adds.

- ② $(-6)(-4) = 24$, and $24 \div 8 = 3$.
Two negatives make a positive.

- ③ $11 + 3 = 14$.
Multiplication before addition ✓

ANSWER

14

EXAMPLE 2 Solve $5x - 4 = 3x + 10$ and check the answer.

- ① Subtract $3x$: $2x - 4 = 10$.

- ② Add 4: $2x = 14$, so $x = 7$.

- ③ Check: $35 - 4 = 31$ and $21 + 10 = 31$ ✓

ANSWER

$x = 7$

EXAMPLE 3 Find the n th term of 2, 9, 16, 23, ... and its 100th term.

- ① $d = 7$, so start from $7n$.
- ② $7n$ gives 7 at $n = 1$, but the term is 2: subtract 5.
- ③ $7n - 5$; at $n = 100$, 695.
Check $n = 2$: 9 ✓

ANSWER

$7n - 5$ and 695

03 Interactive model

Run the revision as a relay: one section per group, each producing three questions and their answers for the others. Writing the questions revises more than answering them.

Number and algebra in eight questions

$-5 - 4$ equals:

-1

-9

1

9

$(-6)(-4)$ equals:

-24

24

-10

10

$\frac{2}{3} \div \frac{4}{9}$ equals:

$\frac{8}{27}$

$1\frac{1}{2}$

$\frac{2}{3}$

6

$2\frac{3}{4} + 1\frac{5}{6}$ equals:

$$3\frac{8}{10}$$

$$4\frac{7}{12}$$

$$4\frac{1}{2}$$

$$3\frac{19}{12}$$

$2(3x - 4)$ equals:

$$6x - 4$$

$$6x - 8$$

$$5x - 8$$

$$6x + 8$$

$3x + 7 = 22$ gives:

$$x = 5$$

$$x = 7$$

$$x = 15$$

$$x = 9$$

$6.3 \div 0.7$ equals:

0.9

9

90

0.09

The n th term of $5, 8, 11, 14$ is:

$3n$

$3n + 2$

$5n$

$n + 3$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Directed number	Musbat va manfiy son	Целое число со знаком
Reciprocal	Teskari son	Обратное число
Like terms	O'xshash hadlar	Подобные слагаемые
Equation	Tenglama	Уравнение
Lowest common multiple	EKUK	НОК
Highest common factor	EKUB	НОД
Square root	Kvadrat ildiz	Квадратный корень
nth term	n-chi had	n-й член

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Subtracting a negative:

subtracts twice

adds

gives zero

changes nothing

Dividing by a fraction is the same as:

multiplying by it

multiplying by its reciprocal

adding it

subtracting it

Like terms can be:

multiplied only

collected

ignored

squared

A solution should always be:

rounded

checked in the original equation

doubled

left as a fraction

0.8×0.04 has how many decimal places?

1

2

3

4

In $dn + (a - d)$ the d is:

the first term

the common difference

the position

the answer

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-7 + 3$

ANSWER -4

2. $-5 - 4$

ANSWER -9

3. $(-6)(-4)$

ANSWER 24

4. Simplify $3x + 5x - 2x$

ANSWER $6x$

5. Solve $3x + 7 = 22$

ANSWER $x = 5$

6. $\sqrt{144}$

ANSWER 12

7. 3.6×10^3

ANSWER 3600 **Medium · the standard question**

7 PROBLEMS

1. $3 - (-8)$

ANSWER 11

2. $\frac{2}{3} \div \frac{4}{9}$

ANSWER $1\frac{1}{2}$

3. Expand $2(3x - 4)$

ANSWER $6x - 8$

4. Solve $5x - 4 = 3x + 10$

ANSWER $x = 7$

5. $HCF(12, 18)$

ANSWER 6

6. $LCM(12, 18)$

ANSWER 36

7. $6.3 \div 0.7$

ANSWER 9

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{3}{4} + 1\frac{5}{6}$

ANSWER $4\frac{7}{12}$

2. $4\frac{1}{3} - 1\frac{5}{6}$

ANSWER $2\frac{1}{2}$

3. Solve $2(x - 3) = 14$

ANSWER $x = 10$

4. The n th term of 5, 8, 11, 14

ANSWER $3n + 2$

5. The 100th term of $4n - 1$

ANSWER 399

6. 3.847 to 2 d.p.

ANSWER 3.85

7. 0.8×0.04

ANSWER 0.032

07 Homework

Homework — 5 tasks

Show the working line by line; the marks in the annual paper are for the method as much as the answer.

1. Work out $-9 + 4 - (-6)$ and $(-8)(-3) \div 4$.

2. Work out $3\frac{1}{2} + 2\frac{2}{3}$ and $5\frac{1}{4} - 2\frac{2}{3}$.

3. Solve $4x - 5 = 2x + 11$ and check your answer.

4. Find the n th term of $4, 11, 18, 25, \dots$ and its 50th term.
5. Work out 0.9×0.06 , $4.8 \div 0.6$ and 2.5×10^3 .